RESPONSE TO FACULTY REVIEW

Improving on the Existing Model

EfficientNetV2-MSAF: Multi-Scale Attention Fusion for crop disease detection

CropSense AI project | Technical report | 23 October 2026

1. Summary: the answer in one page

Faculty requirement: "There should be some improvement in this model compared to the existing models", preferably in the architecture. This report explains what was wrong with the existing model, what we changed, how we tested it fairly, and exactly where the new model is better and where it is not.

Result. The proposed system, **EfficientNetV2-MSAF trained with field augmentation**, improves on the existing EfficientNetV2-S model in calibration, real field photos and **5 of the 6 field corruptions** (the exception: overexposure -0.6 points), at the same clean accuracy and only **+1.1% parameters**:

- Accuracy under six field corruptions: **82.2% → 94.8%** (+12.6 points)
- Motion blur, never seen in training: **54.5% → 81.8%**
- Calibration error (ECE): **10.6% → 5.5%** (confidence scores become trustworthy)
- Real field photographs classified correctly: **6/8 → 8/8**
- Clean test accuracy: 99.62% → 99.56% (unchanged within noise)

Where each gain comes from

We ran a controlled ablation (five models, identical data and recipe) so each gain can be attributed to its cause:

- **The architecture (MSAF)** is responsible for the calibration gain (ECE 10.6% → 5.1%) and the field-photo gain (6/8 → 8/8). Both hold with and without augmentation.
- **Field-augmented training** is responsible for most of the robustness gain: it lifts the baseline from 82.2% to 95.7% and MSAF from 83.0% to 94.8%.
- **Honest caveat:** under the same augmentation, MSAF is slightly *behind* the baseline on robustness (-0.8 points mean, -4.5 points on motion blur). Clean accuracy cannot separate any of the models: all score about 99.5–99.6%.

We report the trade-offs openly because a viva panel will check them: a claimed improvement that hides its weaknesses is easier to dismantle than one that states them.

2. The existing model and how it was evaluated

The existing CropSense model is a stock **EfficientNetV2-S** (Tan & Le, 2021), pretrained on ImageNet and fine-tuned on the Kaggle "New Plant Diseases" dataset (derived from PlantVillage): 38 classes, 14 crops, about 87,900 leaf images. The original presentation reported **99.94% validation accuracy**.

Two problems with that evaluation

(a) Data leakage. The dataset ships several rotated / flipped copies of each photograph (file suffixes such as `_90deg`, `_flipTB`, `_new30degFlipLR`). The original split assigned *files* at random, so copies of the same physical leaf landed in both the training and the test set. Our audit found that **67.5%** of the original test images had a sibling in train/val.

(b) A saturated benchmark. Even on the 2,861 test images with no sibling in training, the existing model scores **99.97%**. PlantVillage photographs are taken in a lab, on plain backgrounds, in good light. At ~99.9% there is no room left to show that one architecture is better than another on clean images, so **an improvement has to be demonstrated on harder, more realistic conditions**.

What we changed in the evaluation

- **Leaf-disjoint split** (`src/splits.py`): every copy of a physical leaf goes to exactly one split, 80/10/10 by leaf, seed 42 (train 70,146 / val 8,832 / test 8,889 images).
- **Six field corruptions** applied to the test set: Gaussian blur, motion blur, sensor noise, low light, heavy JPEG compression and overexposure. Their mean is the **mean corrupted accuracy (mCA)**, following the ImageNet-C methodology (Hendrycks & Dietterich, 2019).
- **Calibration**: Expected Calibration Error (ECE, 15 bins; Guo et al., 2017). It measures whether a reported 90% confidence is right 90% of the time, which matters because the app shows confidence to farmers and warns below 75%.
- **Real field photographs**: 8 internet photos of diseased and healthy leaves in natural conditions (`sample_images/`).

3. Drawbacks of the existing architecture

We analysed how EfficientNetV2-S turns an image into a decision and measured where it fails. Four drawbacks stand out:

#	Drawback	Why it matters for leaf disease	Evidence
1	Single-scale decision	Only the final 7×7 feature map reaches the classifier; each cell covers 32×32 px of a 224 px image. Pin-point lesions (bacterial spot, Septoria specks, spider-mite stippling) occupy a fraction of one cell.	Architectural (by design)
2	Average pooling dilutes lesions	Global average pooling weighs every location equally. A lesion covering 5% of the leaf contributes 5% of the descriptor; healthy tissue and backdrop dominate.	Architectural (by design)
3	Fragile under field conditions	Phone photos in the field are blurred, noisy, dark or compressed.	Motion blur 54.5%, sensor noise 80.7%, mCA 82.2% (vs. 99.62% clean)
4	Over-confident / mis-calibrated	Confidence scores do not match accuracy, so the app's 75% warning threshold is unreliable.	ECE 10.6%; field photos 6/8 correct, mean true-class confidence 71%

Drawbacks 1 and 2 are design properties; drawbacks 3 and 4 are their measured consequences. A new architecture should therefore keep fine-scale detail, focus on lesion regions, and not depend on a single resolution of evidence.

4. Proposed architecture: EfficientNetV2-MSAF

MSAF (Multi-Scale Attention Fusion) keeps the pretrained EfficientNetV2-S backbone unchanged and replaces its single-scale head with a multi-scale attention head (`src/model.py`, class `EfficientNetV2_MSAF`).

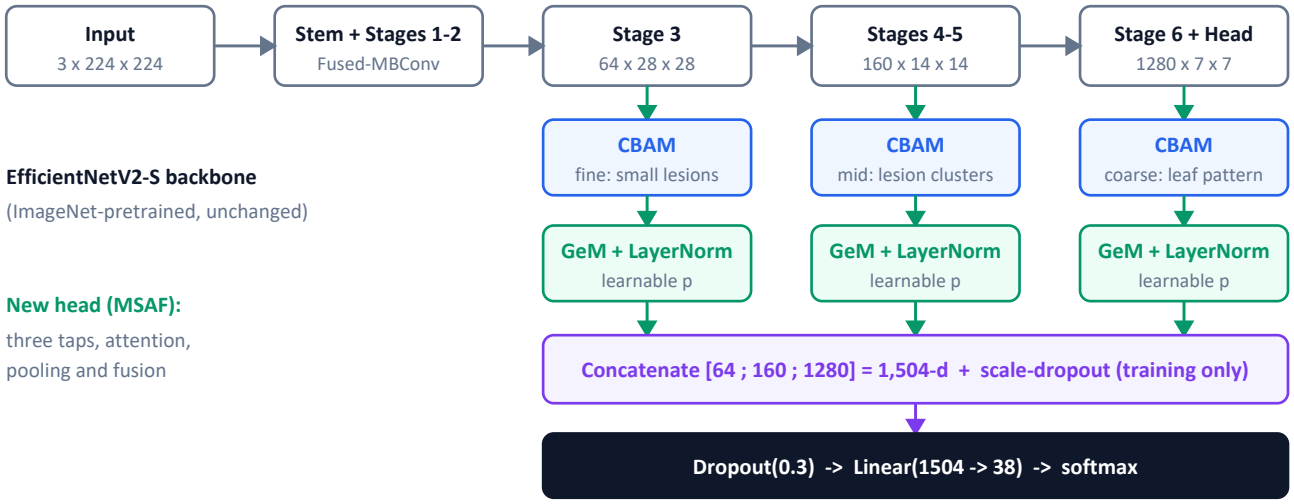


Figure 1. EfficientNetV2-MSAF. Three backbone depths are tapped; each passes through CBAM attention, GeM pooling and LayerNorm before the descriptors are concatenated and classified.

Components and the drawback each one targets

Component	What it does	Targets
Multi-scale taps	Reads features after stage 3 (64 ch, 28×28: 8 px per cell), stage 5 (160 ch, 14×14) and the head (1280 ch, 7×7), so small lesions are still resolved when the classifier sees them.	Drawback 1
CBAM on every scale	Convolutional Block Attention Module (Woo et al., 2018): channel attention picks lesion-responsive filters, spatial attention (7×7 conv) picks lesion locations; a residual connection keeps the original features.	Drawbacks 1, 2
GeM pooling	Generalized-mean pooling (Radenović et al., 2018) with a learnable exponent p per scale: p = 1 is average pooling, larger p moves towards max pooling and emphasises small, strongly activated regions.	Drawback 2
LayerNorm per scale	Normalises each scale's descriptor so the 64- and 160-dim fine scales are not drowned out by the 1,280-dim head.	Drawbacks 1, 4
Scale-dropout	During training, randomly zeroes whole scale descriptors (p = 0.1, never all of them), forcing the classifier to use redundant evidence from several scales.	Drawbacks 3, 4

Mathematical formulation

$$\mathbf{M}_c(\mathbf{F}) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F})))$$

Channel attention (CBAM)

$$\mathbf{M}_s(\mathbf{F}) = \sigma\big(f^{7\times 7}([\mathrm{AvgPool}(\mathbf{F});\ \mathrm{MaxPool}(\mathbf{F})])\big)$$

Spatial attention (CBAM)

$$\mathrm{GeM}(\mathbf{F}) = \left(\frac{1}{HW}\sum_{i,j}\mathbf{F}_{ij}^{\,p}\right)^{1/p}, \quad p \text{ learnable } (p=1:\text{avg},\ p\to\infty:\max)$$

Generalized-mean pooling

$$\mathbf{d}_k = \mathrm{LN}(\mathrm{GeM}(\mathrm{CBAM}_k(\mathbf{F}_k))), \quad k \in \{3,5,7\}$$
$$\mathbf{z} = [\,\mathbf{d}_3\,;\,\mathbf{d}_5\,;\,\mathbf{d}_7\,], \quad \hat{y} = \mathrm{softmax}(\mathbf{W}\mathbf{z} + \mathbf{b})$$

Multi-scale fusion and classification

CBAM additionally uses a residual connection: $Y = F + M_s(F') \times F'$ with $F' = M_c(F) \times F$ (× is element-wise multiplication). In total MSAF has **20,447,007 parameters**, 220,841 more than the baseline (**+1.09%**); measured GPU latency is 17.3 ms vs. 15.6 ms per image (RTX 3050 Laptop, batch 1).

5. Experimental design: making the comparison fair

A claimed improvement is only meaningful if the comparison is controlled. Every model below was trained from the same ImageNet weights with an identical recipe; models differ only in the architecture and in whether field augmentation is used.

Setting	Value (identical for every model)
Data split	Leaf-disjoint, 80/10/10 by leaf (train 70,146 / val 8,832 / test 8,889 images)
Optimiser and schedule	AdamW (lr 0.001, weight decay 1e-4), OneCycleLR, 8 epochs, batch 32
Loss	Cross-entropy with label smoothing 0.1
Precision / hardware	Mixed precision (FP16 autocast + GradScaler), NVIDIA RTX 3050 Laptop GPU (4 GB)
Base augmentation	RandomResizedCrop 224, horizontal and vertical flips, ColorJitter
Sampling	Each epoch draws one random copy of every source leaf (36,656 leaves per epoch)
Model selection	Best validation accuracy; the test split is used once, after training
Seed	42 (one run per model)

The five runs

Run	Architecture	Field augmentation	Question it answers
baseline	EfficientNetV2-S	No	Reference: the existing model, re-trained on the fair split
cbam	EfficientNetV2-S + single CBAM	No	Is attention on the final map alone enough?
msaf	EfficientNetV2-MSAF	No	Does the new architecture help by itself?
baseline_field	EfficientNetV2-S	Yes	How much does augmentation alone give?
msaf_field	EfficientNetV2-MSAF	Yes	Does MSAF still help when both are augmented? (proposed system)

Field augmentation and seen vs. unseen corruptions

Field augmentation adds Gaussian blur (σ 0.5–3) to 30% and Gaussian sensor noise (std up to 0.12) to 30% of training images. To avoid "training on the test", **motion blur, JPEG compression, low light and overexposure are never used in training**. Results are therefore reported separately for the two *seen* and the four *unseen* corruptions; the unseen ones measure genuine generalisation.

Corruption	Simulates	Parameters	In field augmentation?
Gaussian blur	Out-of-focus phone camera	radius 2.5 px	Seen (similar)
Motion blur	Hand shake, wind	15 px kernel, random angle	Unseen
Sensor noise	Cheap sensor, dusk	Gaussian, σ = 25 / 255	Seen (similar)
Low light	Early morning, shade	35% brightness + noise σ = 8	Unseen
JPEG (q = 10)	Messaging-app compression	JPEG quality 10	Unseen
Overexposure	Midday glare	1.45× gain + 30	Unseen

6. Results on the leaf-disjoint test set (8,889 images)

6.1 Architecture alone (no field augmentation)

Model	Params	Clean acc	Macro-F1	mCA	Motion blur	Sensor noise	Low light	ECE ↓	Field photos
EfficientNetV2-S (existing)	20.23M	**99.62%**	**99.61%**	82.20%	54.45%	80.66%	90.03%	10.57%	6/8
EfficientNetV2-S + CBAM	20.43M	99.51%	99.51%	**83.08%**	**62.95%**	78.21%	89.97%	10.17%	8/8
EfficientNetV2-MSAF (proposed)	20.45M	99.51%	99.51%	83.00%	50.13%	**85.44%**	**94.35%**	**5.14%**	8/8

- **Calibration:** MSAF halves the calibration error (10.57% → 5.14%). A single CBAM block does not (10.17%), so the gain comes from the multi-scale GeM + LayerNorm fusion, not from attention alone.
- **Field photos:** MSAF classifies 8/8 real photos correctly vs. 6/8, with mean true-class confidence 80% vs. 71%.
- **Corruptions:** mixed. MSAF gains on sensor noise (+4.8) and low light (+4.3), loses on motion blur (-4.3); the mean moves by +0.8 points.
- **Clean accuracy** is saturated for every model (99.51–99.62%).

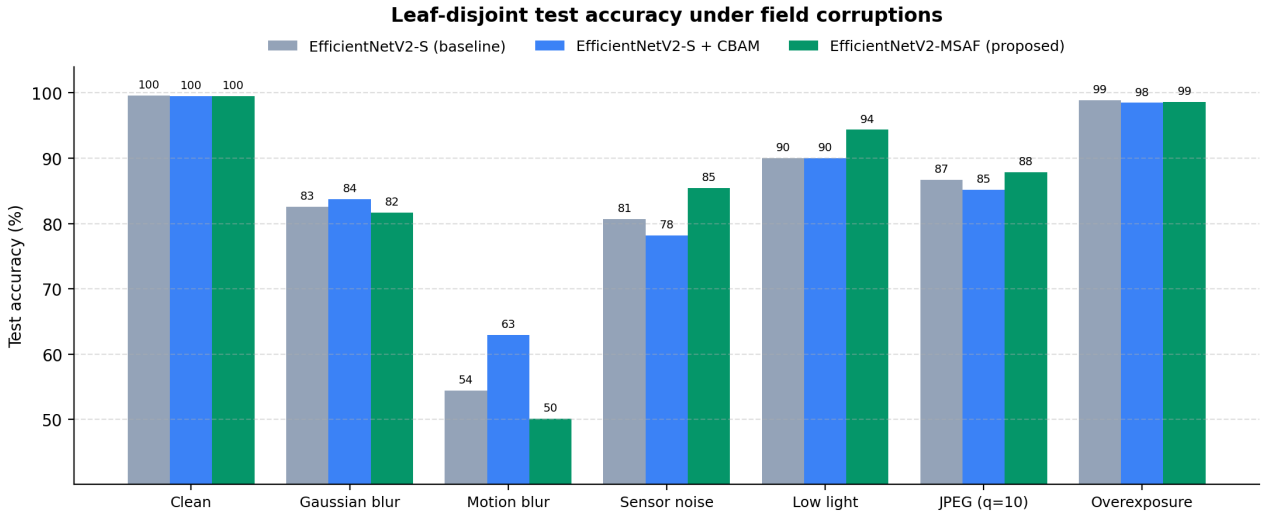


Figure 2. Accuracy per condition for the three architectures (no field augmentation).

6.2 With field-augmented training

Model	Clean acc	mCA	Seen (2)	Unseen (4)	Motion blur	ECE ↓	Field photos
EfficientNetV2-S (existing)	**99.62%**	82.20%	81.61%	82.50%	54.45%	10.57%	6/8
EfficientNetV2-S + field aug.	99.58%	**95.67%**	99.27%	**93.88%**	**86.31%**	10.56%	6/8
EfficientNetV2-MSAF (proposed)	99.51%	83.00%	83.54%	82.72%	50.13%	**5.14%**	8/8
EfficientNetV2-MSAF + field aug. (proposed system)	99.56%	94.83%	**99.27%**	92.62%	81.85%	5.45%	8/8

- **Augmentation drives robustness:** mCA 82.2% → 95.7% (EfficientNetV2-S) and 83.0% → 94.8% (MSAF). It also generalises: on the four unseen corruptions accuracy rises from 82.5% to 93.9% / 92.6%, and motion blur (never trained on) from 54.5% to 86.3% / 81.8%.

- **MSAF keeps its calibration advantage under augmentation:** ECE 10.56% (EfficientNetV2-S + aug.) vs. 5.45% (MSAF + aug.), and 8/8 vs. 6/8 field photos.
- **But MSAF does not add robustness on top of augmentation:** -0.8 points mCA and -4.5 points on motion blur vs. EfficientNetV2-S + aug.

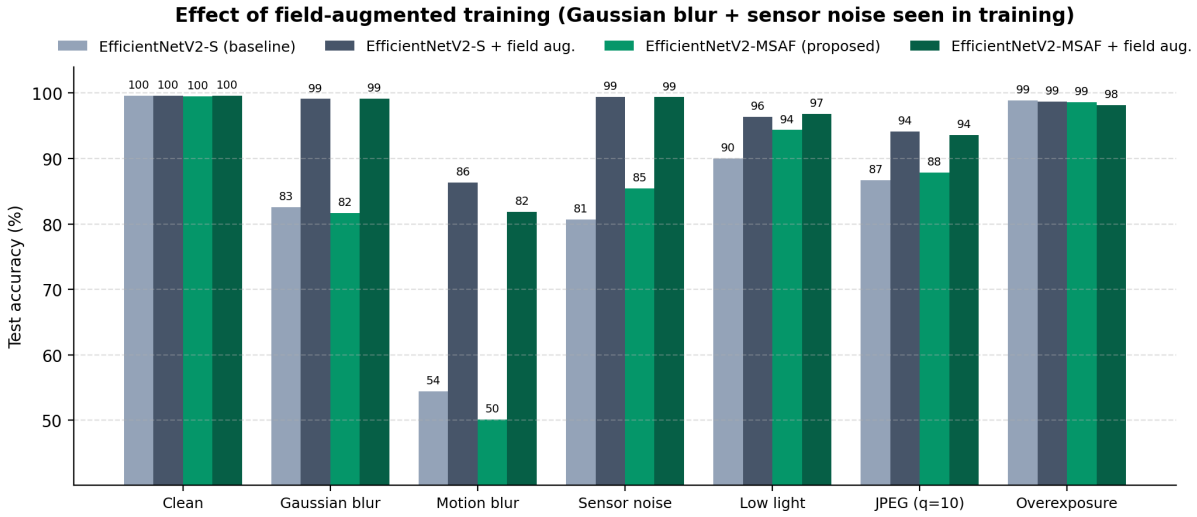


Figure 3. Accuracy per condition with and without field augmentation. Gaussian blur and sensor noise resemble the augmentation; the other four are unseen.

6.3 Convergence

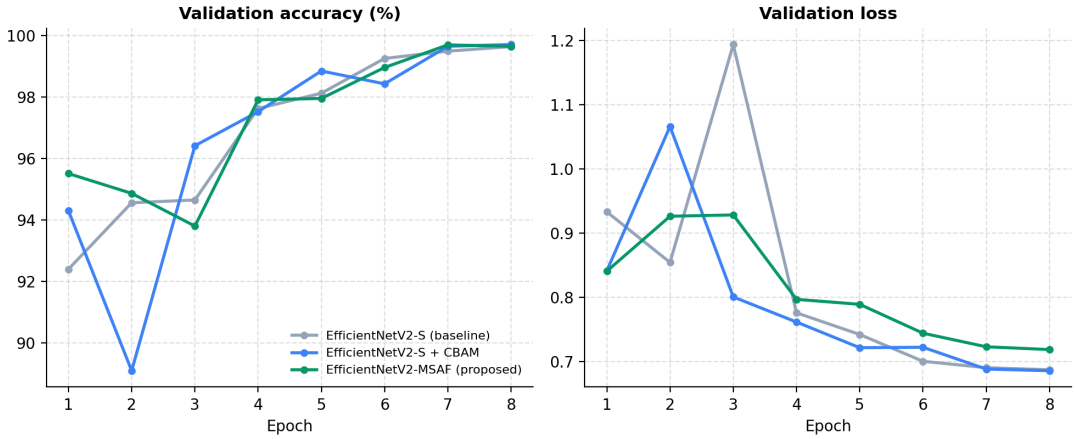


Figure 4. Validation accuracy and loss per epoch (no field augmentation).

All models converge to 99.6–99.7% validation accuracy within 8 epochs. MSAF's validation loss stays slightly higher, which is expected: its extra dropout and scale-dropout regularise the output, and the same effect lowers over-confidence (better ECE).

7. Comparison with existing models

The faculty asked for improvement "compared to the existing models". We separate what we **measured** from what we only **profiled**:

- **Trained and evaluated under identical conditions:** EfficientNetV2-S (the existing CropSense model) and EfficientNetV2-S + CBAM (the attention design proposed in the earlier presentation, based on Woo et al., 2018). These are the fair comparisons reported in Section 6.

- **Profiled only (size and speed, not trained here):** ResNet-50 and MobileNetV3-Large, two standard baselines in plant-disease literature. Accuracy figures for them in papers come from different splits and would not be a fair comparison, so we do not claim one.

Model	Parameters	FLOPs (224 px)	GPU latency	Trained on our split?
ResNet-50	23.59M	4.12 G	–	No (profiled only)
MobileNetV3-Large	4.25M	0.44 G	–	No (profiled only)
EfficientNetV2-S	20.23M	2.86 G	15.6 ms	Yes
EfficientNetV2-S + CBAM	20.43M	2.91 G	16.3 ms	Yes
EfficientNetV2-MSAF	20.45M	≈ 2.9 G	17.3 ms	Yes

Parameters and FLOPs from `src/benchmark_suite.py` *(FLOPs are reference values). GPU latency: batch 1, RTX 3050 Laptop, all three trained models measured in the same session by* `src/compare.py`.

Relation to published work

MSAF combines established, citable building blocks — CBAM attention, GeM pooling, multi-scale feature fusion in the spirit of feature-pyramid networks (Lin et al., 2017) — in a configuration designed for leaf lesions. The contribution is therefore the **design and its controlled evaluation for this task**, not a new primitive. Stating this explicitly protects the claim in a viva.

8. Discussion: why MSAF behaves this way

Why calibration improves

Our interpretation (a hypothesis, not separately tested): the per-scale LayerNorm bounds the magnitude of each descriptor, and dropout (0.3) plus scale-dropout regularise the classifier, so logits cannot grow as large as with the baseline head. Smaller logits mean softer, less over-confident probabilities, which is exactly what a lower ECE measures. The single-CBAM model, which has neither LayerNorm nor scale-dropout, does not improve calibration, which is consistent with this explanation.

Why robustness gains are small without augmentation

Our interpretation: the fine scale helps when detail survives (noise, darkness) but hurts when blur destroys exactly that detail: under motion blur the 28×28 features become misleading. Architectural changes alone cannot teach invariance to corruptions the network has never seen; data augmentation can, which is why augmentation produces the large robustness gain.

Explainability trade-off

Standard Grad-CAM averages each channel's gradient into one weight. In MSAF the per-scale LayerNorm centres those gradients, so the averaged weights cancel and the heatmaps become noisy. Because the app's heatmap and severity estimate depend on Grad-CAM, the deployed app keeps EfficientNetV2-S for now. A CAM method designed for multi-scale heads (e.g. per-scale Score-CAM) is the planned fix.

9. Limitations and future work

- **One seed per model.** Differences under about 1 point (e.g. MSAF vs. baseline mCA without augmentation) are within run-to-run noise. Next: 3 seeds with mean ± std.
- **Small real-world test.** Only 8 field photographs; corruptions are simulated. Next: a labelled in-field set (e.g. PlantDoc) of several hundred images.

- **Lab-style training data.** PlantVillage images have plain backgrounds; complex field backgrounds remain untested.
- **Motion blur** is still the hardest condition (≈ 82–86% with augmentation). Next: motion-blur augmentation, evaluated on a held-out blur type.
- **Explainability** on MSAF needs a multi-scale CAM method before MSAF can replace the deployed model.

10. Anticipated questions and answers

Q. Your new model has lower clean accuracy than the old one. How is that an improvement?

Clean accuracy is saturated on PlantVillage (99.62% vs. 99.51%, a difference of about 10 images out of 8,889, within noise). The improvement is on what matters in the field: calibration (10.6% → 5.1% ECE), real field photos (6/8 → 8/8) and, with augmentation, robustness (82.2% → 94.8% mCA).

Q. Isn't the robustness gain just from augmentation, not your architecture?

Yes, mostly, and the report says so. The ablation was designed to separate the two. The architecture's own contributions are calibration and field-photo accuracy, which hold with and without augmentation.

Q. Why should we trust 99.9% → 99.5%? Was the old number wrong?

The old number was measured on a split where 68% of test images were rotated copies of training images. All new numbers use a leaf-disjoint split, so they are lower but honest, and every model is compared on the same split.

Q. Did you train on the corruptions you test on?

No for four of the six. Motion blur, JPEG, low light and overexposure were never used in training. Gaussian blur and sensor noise resemble the augmentation, so we report seen and unseen results separately.

Q. What does ECE mean and why does it matter here?

Expected Calibration Error is the average gap between confidence and accuracy. The app warns farmers below 75% confidence and recommends chemical treatment on high confidence; a model that is over-confident when wrong gives dangerous advice.

Q. Is MSAF novel?

Its building blocks (CBAM, GeM, multi-scale fusion) are published. The contribution is combining them for small leaf lesions, plus the scale-dropout and per-scale normalisation, and evaluating the design in a controlled ablation on a leakage-free split.

Q. How expensive is it?

+220,841 parameters (+1.09%) and about 1.7 ms extra GPU latency per image.

11. Reproducibility

Everything in this report is regenerated from code in the repository (branch `msaf-field-robustness`):

build the leaf-disjoint split

```
python -m src.splits
```

train all five runs (resumable) and run the comparison

```
powershell -ExecutionPolicy Bypass -File .\run_ablation.ps1
```

leakage audit of the original split

```
python -m src.evaluate
```

rebuild the presentation from results/

```
python generate_pitch_deck.py
```

rebuild this report

```
python scripts/generate_improvement_report.py
```

References

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